ProprioVLA: Self-Supervised Proprioceptive Grounding for Vision-Language-Action Policies

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Abstract

Vision-language-action (VLA) policies have shown strong promise as generalist controllers for robot manipulation, yet their action predictions can remain fragile under changes in visual perspective. We identify a systematic grounding failure: lateral camera shifts induce co-directional shifts in predicted robot trajectories, suggesting that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. To bridge this gap, we introduce PROPRIOVLA, a self-supervised learning framework that incentivizes self-grounding in VLA policies. PROPRIOVLA guides the policy to visually attend to the robot itself with learned grounding queries, based on the key insight that *state-aligned residual flow provides implicit supervision for self-grounding*. Specifically, PROPRIOVLA first mines confidence-weighted self-grounding heatmaps from robot interaction by selecting residual motion that is salient in the image and aligned with proprioceptive change. It then trains the learned-query attention with a confidence-weighted normalized KL objective, matching the attention distribution to these heatmaps while retaining the standard action-supervised VLA objective. This yields self-supervised proprioceptive grounding without manual annotations, external grounding datasets, or stronger foundation-model supervision. Extensive experiments on both real-robot and simulation environments show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, suggesting that proprioceptive grounding is a practical route toward spatially reliable VLA control.

1 Introduction

Vision-language-action (VLA) policies make language-conditioned robot manipulation increasingly general, but reliable control requires spatial grounding in the robot's own embodiment. Recent robot foundation models show that broad pretraining and large robot datasets can yield policies that transfer across tasks and embodiments [2, 3, 7, 5]. For manipulation, however, recognizing the instructed object is not sufficient: the policy must relate visual evidence to the robot's proprioceptive state and to the geometry of the surrounding scene. Spatial robustness under changes in visual perspective is therefore a core requirement for dependable VLA control.

A key unresolved failure mode is that a VLA policy can be task-grounded without being self-grounded. In controlled camera-perturbation evaluations where the underlying robot-scene state is unchanged, lateral camera shifts induce co-directional shifts in predicted robot trajectories. This systematic behavior suggests that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, instead of anchoring actions to proprioceptive state and robot-object geometry. A policy trained only to match demonstrations under a fixed observation distribution can therefore appear competent while remaining brittle when the same physical state is viewed from a different perspective.

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Existing grounding-oriented VLA methods show that perception, attention allocation, and embodiment cues must be treated as first-class control problems. Recent methods improve spatial awareness with visual trajectory prompts, 3D representations, camera-space action formulations, view synthesis, or explicit treatments of proprioception [11, 8, 9, 4, 10, 6]. These approaches are valuable, but they do not directly provide a self-supervised signal that tells the policy where its own acting body appears in the image. For spatially reliable manipulation, the missing cue is not only object grounding or view robustness, but a binding between proprioceptive state and visual evidence of the robot itself.

PROPRIOVLA addresses this missing-self problem by learning self-supervised proprioceptive grounding from robot interaction. The key insight is that the robot reveals itself through visual motion synchronized with changes in proprioceptive state. Rather than using proprioception as a passive input token, PROPRIOVLA uses proprioceptive change to derive visual supervision from state-aligned residual flow and uses this supervision to train learned grounding queries. This signal highlights residual motion that is both locally salient in the image and aligned with proprioceptive change, indicating where the robot's own visual evidence appears during interaction.

The method realizes this idea through confidence-weighted self-grounding heatmaps and a normalized KL grounding objective. First, PROPRIOVLA mines heatmaps from robot interaction by selecting residual motion that is salient in the image and aligned with proprioceptive change, while down-weighting uncertain frames. These heatmaps supervise a lightweight learned-query adapter over visual tokens, encouraging the policy representation to localize the visible robot body without manual masks, external grounding datasets, or supervision from stronger foundation models.

Experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. We evaluate PROPRIOVLA in real-robot and simulation environments, including controlled camera-perturbation settings that directly probe view robustness. Across these evaluations, PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation. Ablations further support the role of the learned-query grounding path: state-aligned heatmaps provide spatial supervision, and normalized KL supervision stabilizes attention learning across target scales. These results suggest that self-supervised proprioceptive grounding is a practical route toward spatially reliable VLA control.

Our contributions are:

- We identify a systematic grounding failure in VLA policies: camera shifts can induce co-directional shifts in predicted robot trajectories, revealing reliance on image-space shortcuts rather than proprioceptive grounding.
- We introduce state-aligned residual flow as an implicit supervision signal for self-grounding, enabling confidence-weighted heatmaps to be mined from robot interaction without manual annotations, external grounding datasets, or stronger foundation-model supervision.
- We propose PROPRIOVLA, a self-supervised learning framework that trains learned grounding queries with normalized KL supervision to improve task success and robustness in real-robot and simulation environments.

2 Related Work

Vision-language-action models. Vision-language-action (VLA) policies scale language-conditioned robot control by coupling vision-language backbones with robot action prediction. RT-1 and RT-2 established the value of large robot datasets and web-scale semantic priors [2, 3]; recent open and generalist models further improve transfer across tasks, embodiments, and action parameterizations [7, 5, 1]. However, this line of work mainly optimizes task success under the training observation distribution. It does not directly test whether the action-relevant visual evidence is anchored to camera-invariant robot-object geometry. PROPRIOVLA studies this missing failure mode: under camera perturbations, a fine-tuned VLA can follow image-coordinate shortcuts instead of grounding actions in the robot's own state.

Spatial and proprioceptive grounding in VLA policies. Recent work improves VLA perception by making space, viewpoint, or proprioception more explicit. TraceVLA adds visual trajectory prompts to increase spatial-temporal awareness [11]; SpatialVLA injects 3D spatial representations [8]; OC-VLA predicts actions in camera space to reduce observation-action mismatch [9]; and VistaBot uses

geometry-aware view synthesis for cross-view robustness [4]. These methods show that perception is a central bottleneck, but they depend on explicit prompts, 3D encodings, camera coordinate transforms, or generative view modules. A complementary line questions proprioception itself: State-free Policy removes proprioceptive inputs to avoid trajectory overfitting [10], while GAP shows that proprioceptive gradients can suppress visual learning during motion transitions [6]. PROPRIOVLA takes a different stance: it does not discard proprioception or treat it as a passive token. Instead, it uses proprioceptive change to mine state-aligned residual-flow heatmaps, turning the robot's own motion into self-supervision for learned visual attention.

3 Does a VLA Know Where It Is?

A manipulation policy must know where it is, not only what object the language instruction names. We probe this self-location requirement at two levels. The action-level probe asks whether a predicted trajectory remains tied to the fixed robot-scene state when only the camera view is shifted. The representation-level probe asks whether robot-relevant attention localizes the arm, gripper, or end-effector rather than only task objects or background context. Together, these probes expose a *missing-self* failure mode: a policy can appear task-grounded while lacking a stable visual grounding of the body that acts.

3.1 Finding 1: The Action Frame Is Entangled with the Image Frame

We diagnose action-frame grounding by perturbing the view while holding the physical manipulation state fixed. Given observation o_t, instruction x, proprioceptive state s_t, and policy π_θ, let $\tau_\theta = \pi_\theta(o_t, x, s_t)$ be the predicted trajectory from the original view. We then apply a lateral camera or image shift $T_{\Delta v_{\mathrm{cam}}}$ and compute $\tau'_\theta = \pi_\theta(T_{\Delta v_{\mathrm{cam}}}(o_t), x, s_t)$. The instruction, robot state, object configuration, and physical scene are unchanged; only the image frame moves.

If the action frame were grounded in robot-scene geometry, the prediction would remain stable under this camera-only perturbation, up to the known transform used for visualization. A missing-self failure appears when the predicted end-effector displacement follows the image displacement. We quantify this effect after projecting the trajectory change onto a common lateral coordinate system:

$$\mathrm{TSR} = \frac{\|\Delta\tau_{\mathrm{pred}}\|_2}{\|\Delta v_{\mathrm{cam}}\|_2 + \epsilon},$$
$$\mathrm{SDA} = \frac{\langle \Delta\tau_{\mathrm{pred}}, \Delta v_{\mathrm{cam}} \rangle}{\|\Delta\tau_{\mathrm{pred}}\|_2 \|\Delta v_{\mathrm{cam}}\|_2 + \epsilon}.$$

High TSR means the prediction changes strongly under a view-only shift; high SDA means the change is co-directional with the image shift. The combination indicates that the learned action frame is partially entangled with the image frame rather than fully anchored to proprioceptive state and robot-object geometry.

3.2 Finding 2: Task Grounding Does Not Imply Self-Grounding

We next test whether action-relevant representations localize the acting body. For a VLA or its vision-language backbone, we use robot-centric prompts such as "Where is the robot arm?", "Please identify the robot itself in the scene," and "Where is the manipulator?" We then visualize attention heatmaps, attention rollout, Grad-CAM maps, cross-attention maps, or token relevance maps. The same analysis can be applied to policy tokens used for action prediction, comparing task relevance with self relevance.

This probe separates task grounding from self-grounding. A model may attend to task objects, the tabletop, or salient background regions while assigning little relevance to the robot arm, gripper, or end-effector. Such attention can support object recognition and instruction following, but it does not bind the action to the body that will execute it. Robust VLA control therefore requires binding proprioceptive state to visual evidence of the acting body, rather than treating proprioception as a passive input that the visual pathway can ignore.

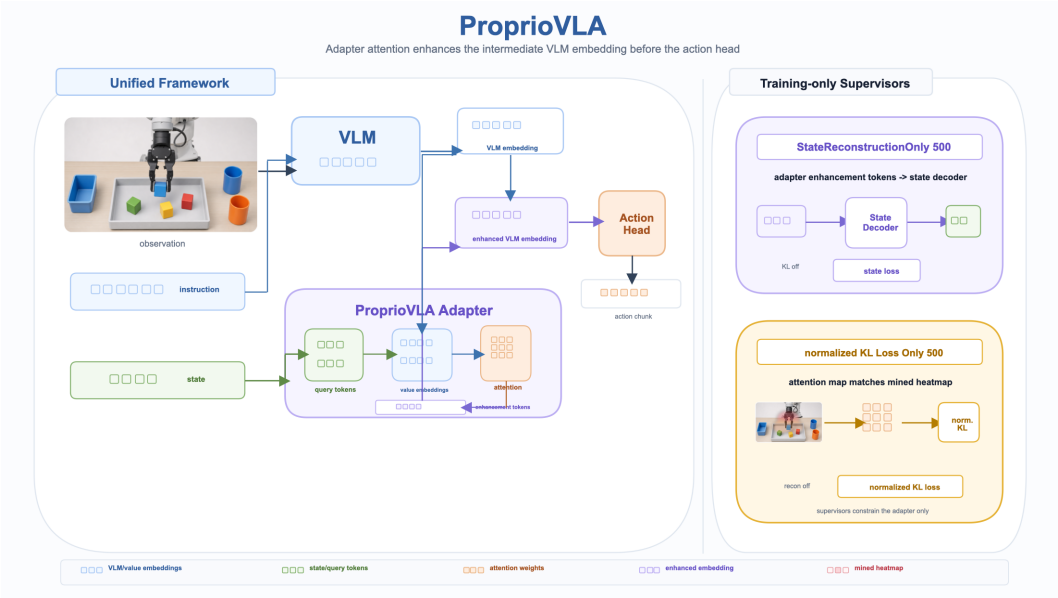

Figure 1: Overview of PROPRIOVLA. Given an observation, instruction, and proprioceptive state, the base VLA produces a standard multimodal representation. PROPRIOVLA adds a lightweight learned-query grounding adapter over visual tokens and supervises its attention with mined self-grounding heatmaps. The heatmap supervision is used only during training.

4 PROPRIOVLA

PROPRIOVLA adds a proprioceptive grounding path to a VLA policy while leaving the standard action-supervised objective intact. During training, the method mines self-grounding targets from state-aligned residual motion and uses them to supervise learned grounding queries over visual features. At test time, the mined heatmaps are not required.

4.1 Self-Grounding from State-Aligned Motion

The robot reveals itself through motion that is synchronized with its state change. Given consecutive observations (o_t, o_{t+1}) and proprioceptive states (s_t, s_{t+1}), we define $\Delta s_t = s_{t+1} - s_t$. We estimate dense visual motion $F_t(p)$ between frames and subtract global or background motion to obtain residual motion $R_t(p)$ at image location p. This residual suppresses camera drift and static-scene variation, emphasizing motion caused by the robot or by objects directly coupled to the robot.

We convert residual motion into a confidence-weighted self-grounding target. Let $\mathrm{sal}(R_t(p))$ denote local residual-motion saliency and let

$$\alpha_t(p) = \left[\cos\big(\psi_R(R_t(p)), \psi_s(\Delta s_t)\big)\right]_+$$

measure state-motion alignment after lightweight projections ψ_R and ψ_s. The unnormalized heatmap score is

$$e_t(p) = \mathrm{sal}(R_t(p))\, \alpha_t(p).$$

We normalize e_t over the image or patch grid to obtain H_t, and set a confidence $c_t \in [0, 1]$ from the total aligned-motion mass. Low confidence downweights frames where the self-motion cue is weak or ambiguous. Thus the training target is mined from interaction alone, without manual masks, external grounding datasets, or stronger foundation-model supervision.

4.2 Learned Queries for Self-Grounding

In PROPRIOVLA, a small set of learned queries forms the grounding path that localizes the robot in the image. Let $Q = \{q_m\}_{m=1}^{M}$ be M learned query latents, and let $K_t, V_t = f_v(o_t)$ be visual keys

and values from the VLA vision encoder. The queries attend over visual patches according to

$$A_t(m,p) = \mathrm{softmax}_p\left(\frac{\bar{q}_m \bar{K}_t(p)^\top}{\tau}\right), \qquad A_t(p) = \frac{1}{M}\sum_{m=1}^{M} A_t(m,p),$$

where $\bar{q}_m$ and $\bar{K}_t(p)$ are ℓ_2-normalized query and key projections, and τ is a temperature. The averaged attention map A_t is the model's predicted self-grounding distribution over visual patches.

4.3 Normalized KL Grounding Objective

The mined heatmap supervises the learned-query attention through a normalized KL objective. Let U_t denote the uniform distribution over valid visual tokens. The grounding loss is

$$\mathcal{L}_{\mathrm{ground}} = c_t \frac{\mathrm{KL}(H_t \,\|\, A_t)}{\mathrm{KL}(H_t \,\|\, U_t) + \epsilon},$$

after resizing H_t to the attention grid and normalizing it over valid tokens. The denominator makes the auxiliary loss comparable across target sharpness and token counts: matching the target approaches zero, while uniform attention is normalized to approximately one. The complete training loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{act}} + \lambda_g \mathcal{L}_{\mathrm{ground}}.$$

Here $\mathcal{L}_{\mathrm{act}}$ is the standard imitation loss used to fine-tune the VLA.

5 Experiments

The experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. The evaluation links three evidence types: task performance, action stability under camera perturbations, and visual grounding of the acting robot.

5.1 Experimental Setup

We evaluate a VLA backbone fine-tuned on language-conditioned manipulation demonstrations in simulation and on a real robot. Each episode contains visual observations o_t, proprioceptive state s_t, language instruction x, and action supervision a_t. For robustness evaluation, we keep the physical robot-scene state, instruction, and proprioceptive state fixed, then apply lateral view shifts of controlled magnitude. This isolates camera-induced changes from changes in the underlying manipulation state.

We compare four policies under the same demonstrations, action labels, and view-shift protocol: Base VLA, action-supervised fine-tuning, VLA with proprioception concatenation, and PROPRIOVLA. We report task success rate, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and a grounding score. The grounding score is computed as either attention-heatmap alignment with H_t or attention mass on robot regions. Lower trajectory shift and direction agreement indicate a more stable action frame; higher grounding score indicates stronger localization of the acting body. Each setup reports the task suite, demonstration count, train/test split, camera shifts, action parameterization, proprioceptive normalization, optimizer, and the grounding loss weight λ_g.

5.2 Simulation Experiments

Simulation provides controlled access to camera shifts, object layouts, and ground-truth robot geometry. We evaluate normal, mild-shift, medium-shift, and large-shift settings. For each setting, we report task success, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and grounding score. Robustness curves plot these metrics against shift magnitude, making the co-movement between predicted actions and image shifts directly visible.

We also evaluate held-out object layouts and camera positions. These splits separate memorization of a fixed training camera from grounding that transfers across views. Qualitative trajectory overlays compare the original prediction with shifted-view predictions after mapping them to the same robot-centric coordinate frame.

5.3 Real-World Experiments

The real-robot evaluation tests whether the same failure mode appears under sensor noise, calibration error, and real scene variation. We evaluate several manipulation tasks under the original view and shifted views, reporting task success, success drop under perturbation, action prediction error on logged demonstrations, and grounding score.

For qualitative analysis, we visualize baseline attention, the mined state-aligned heatmap, PROPRIOVLA attention, and trajectory overlays for the same scene. The comparison checks whether attention remains dominated by task objects and background context or shifts toward the arm, gripper, and end-effector while the trajectory remains stable.

5.4 Ablations

The ablation study isolates the mechanisms that distinguish PROPRIOVLA from simple state conditioning. We evaluate four variants: (1) removing the visual self-grounding loss $\mathcal{L}_{\text{ground}}$; (2) replacing normalized KL with an unnormalized localization loss; (3) using raw optical flow instead of state-aligned residual motion; and (4) replacing the learned grounding queries with state-derived queries. Each ablation is evaluated with the same success, action, trajectory-shift, and grounding metrics.

These ablations test the central mechanism: the policy must use proprioception to mine visual self-grounding targets, rather than merely receive proprioceptive values as extra inputs. The comparison separates three alternatives: gains from additional state, gains from motion saliency alone, and gains from learned-query self-grounding.

6 Limitations

PROPRIOVLA assumes that the robot is at least partially visible during training. If the embodiment is consistently outside the camera view, the state-aligned motion signal cannot provide useful self-grounding supervision, and the visual query has no direct evidence from which to recover the robot's own state.

State-aligned residual motion can be ambiguous when non-robot motion is coupled to the robot. Grasped objects, deformable materials, moving backgrounds, or other agents may move synchronously with proprioceptive change and receive high alignment scores. Confidence weighting reduces weak supervision, but it cannot fully distinguish the robot body from every object moved by that body.

Finally, the evaluation covers a finite set of embodiments, camera perturbations, and manipulation tasks. Broader multi-camera setups, mobile manipulation, multi-robot scenes, long-horizon tasks, and highly dynamic environments remain important tests for whether proprioceptive self-grounding scales beyond controlled perturbation regimes.

7 Conclusion

This paper studies a missing-self failure mode in VLA policies: under camera-only viewpoint shifts, predicted robot trajectories can move with the image frame even when the physical robot-scene state is unchanged. The diagnostic shows that task grounding and self-grounding are distinct requirements for spatially reliable manipulation.

PROPRIOVLA addresses this issue through three connected mechanisms: state-aligned motion supervision, learned grounding queries, and normalized KL attention supervision. The framework mines self-grounding heatmaps from robot interaction without manual masks or external grounding datasets and uses them to select robot-related visual evidence during training. Learning to see the self is a practical step toward VLA policies whose actions are grounded in embodiment rather than in the image frame alone.

References

[1] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0: A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.

[2] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Jasmine Hsu, et al. RT-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.

[3] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, 2023.

[4] Songen Gu, Yuhang Zheng, Weize Li, Yupeng Zheng, Yating Feng, Xiang Li, Yilun Chen, Pengfei Li, and Wenchao Ding. VistaBot: View-robust robot manipulation via spatiotemporal-aware view synthesis. *arXiv preprint arXiv:2604.21914*, 2026.

[5] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[6] Jingxian Lu, Wenke Xia, Yuxuan Wu, Zhiwu Lu, and Di Hu. When would vision-proprioception policies fail in robotic manipulation? *arXiv preprint arXiv:2602.12032*, 2026.

[7] Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, et al. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024.

[8] Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, and Xuelong Li. SpatialVLA: Exploring spatial representations for visual-language-action model. *Robotics: Science and Systems*, 2025.

[9] Tianyi Zhang, Haonan Duan, Haoran Hao, Yu Qiao, Jifeng Dai, and Zhi Hou. Grounding actions in camera space: Observation-centric vision-language-action policy. *arXiv preprint arXiv:2508.13103*, 2025.

[10] Juntu Zhao, Wenbo Lu, Di Zhang, Yufeng Liu, Yushen Liang, Tianluo Zhang, Yifeng Cao, Junyuan Xie, Yingdong Hu, Shengjie Wang, et al. Do you need proprioceptive states in visuomotor policies? *arXiv preprint arXiv:2509.18644*, 2025.

[11] Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. TraceVLA: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. *arXiv preprint arXiv:2412.10345*, 2025.

A Additional Experimental Details

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